

RAMCO INSTITUTE OF TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING
ACADEMIC YEAR 2018-19
ME6811-PROJECT WORK

Sl.No.	Project title	PO's relevance	PSO's relevance	Outcome
New product development				
1.	Design of Desiccation Aquatic Craniate	POs – 1,2,3,4,5,7,9,10,11,12	PSOs – 1,2	Presentation at Conference on Advanced Materials, Robotics & Automation (CAMRA) 2019
2.	Modelling and Analysis of Solar Powered Hybrid Electric Bike with Waste Heat Recovery System	POs – 1,2,3,4,5,7,9,10,11,12	PSOs – 1,2,3	• International Conference on Science and Innovative Engineering 2019, Jawahar Engineering College, Chennai. • Received Rs.12, 500 as financial support for social relevant and innovative project from RIT research council. • Participated in the technical presentation and won first prize in National Level Presentation held at CEG/MIT Anna university Campus.

3.	Performance Enhancement of HC Blend Domestic Refrigerator Using PCM.	POs – 1,2,3,4,5,7,9,10,11,12	PSOs – 1,2,3	• Participated as Special entry in innoWAH'19, an innovation competition organised by IIT-M PALS. • National Conference On “Recent Trends in Mechanical Engineering (RTME 2019)”, Dr. Sivanthi Aditanar College of Engineering, Tiruchendur
4.	Performance Analysis of Closed Loop Adiabatic Dehumidifier at Vacuum Pressure for Drying Herbal Leaves	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2,3	A Closed loop Adiabatic Dehumidifier is fabricated and its performance is analysed.
Projects in Identified researches				
5.	Improving Productivity Of Solar Still Using Agitator And External Condensing Fans To Meet Water Demand	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2,3	• Presentation at Conference Division, Seventh Sense Research Group 2019 • Received partial financial support from RIT Research Council
6.	Investigation Of Surface Properties Of Heat Treated Electrodes Ni-P/Tio2 Nano Particles Coating	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at International Conference on recent developments in Mechanical Engineering 2019 at S.A.Engineering college, Chennai
7.	Analysis Of Mechanical Properties In SMAW Using 316l & 309 Stainless Steel Base Additive Manufacturing	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at National conference on Innovations in engineering, science and technology in Kongunadu

				College of Engineering and Technology at Trichy”
8.	Analysis Of Surface Properties Of Tungsten Carbide Coating Over Austenite(SS316) Using Plasma Spray Process	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at International Conference on recent developments in Mechanical Engineering 2019 at S.A.Engineering college, Chennai
9.	Copper Corrosion Protection By Reinforcing Secondary Particles	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at International conference on Recent Innovation in Mechanical Engineering (ICRIME 2K19) conducted by Francis Xavier Engineering College, Tirunelveli
10.	Performance Study Of Solar Still With Extended Reflector Compared To Conventional Solar Still	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2,3	Presentation at One day National Conference on Challenges and Opportunities for Mechanical Engineers (COME-2019), Kongunadu College of Engineering and Technology, Trichy
11.	Finite Element Simulation And Experimental Studies On GFRP Aircraft Wing Box Structure Embedded With Smart Piezo Electric Material	POs – 1,2,3,4,5,6,9,10,11,12	PSOs – 1,2,4	Presentation at International Conference on Engineering Materials and Processes (ICEMAP 2019) organized by Tagore Engineering College, Chennai
12.	Influence Of Titanium Of Microstructure, Tensile And Tribological Properties Of Aluminium Borate/ Al-Li Composite.	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at International Conference on recent developments in Mechanical Engineering 2019 at

				S.A.Engineering college, Chennai
13.	Estimation Of Mechanical And Machinability Characteristics Of Al7075-Niti Composites Reinforced With Sic Using AWJM	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at International conference on Recent Innovation in Mechanical Engineering (ICRIME 2K19) conducted by Francis Xavier Engineering College, Tirunelveli
14.	Effect Of Niobium On Microstructure And Mechanical Properties Of Magnesium Based Composites	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at International Conference on recent developments in Mechanical Engineering, S.A Engineering College
15.	Investigation Of Deposition Rate And Corrosion Resistance Of Heat Treated Electroless Nip Coating	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at International conference on Recent Innovation in Mechanical Engineering (ICRIME 2K19) conducted by Francis Xavier Engineering College, Tirunelveli
16.	Mechanical And Material Character Improvisation Of Al7075 with NiTi Composites Reinforced with Sic Particulars	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at International conference on Recent Innovation in Mechanical Engineering (ICRIME 2K19) conducted by Francis Xavier Engineering College, Tirunelveli
Projects in specific applications				
17.	Flat Plate Collector Using Heat Pipe With Individual And Common Condenser	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2,3	Presentation at second International conference on emerging trends in

				Engineering and Technology 2019 at SCAD College of Engineering
18.	Optimization Of Closed Loop Adiabatic Dehumidified Herbal Leaves Drier	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2,3	Presentation at International Conference on Engineering and Advancement in Technology 19,Sri Krishna College of Technology,Coimbatore
19.	Design And Wind Load Analysis For Electric Poles	POs – 1,2,3,4,5,6,9,10,12	PSOs – 1,2,4	Presentation at International conference on Recent Innovation in Mechanical Engineering (ICRIME 2K19) conducted by Francis Xavier Engineering College, Tirunelveli
20.	Conversion Of Waste Plastics Into Construction Materials	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at Conference on Advanced Materials, Robotics and Automation CAMRA 2019, Bannari Amman Institute of Technology, Erode
21.	Conversion Of Wood And Rubber Waste Into Composite Material	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at Two- day National Conference on Materials for Sustainable Development (NCMSD 2019)
22.	Design And Analysis Of Two Wheeler Leg Guard Under Different Loading Conditions	POs – 1,2,3,4,5,6,8,9,10,11,12	PSOs – 1,2,4	Presentation at International Conference on Recent Innovation in Mechanical Engineering, Francis Xavier Engineering College.

23.	Prevention Aid For Banana Tree Farming Due To Heavy Wind	POs – 1,2,3,4,5,6,7,9,10,11,12	PSOs – 1,2,4	• "Tech2Farm" 2018 competition winners @ IIT Madras. • ICTACT - Tamil Nadu Student Innovator Award 2018
24.	Design And Development Of Concur Tint Partition Using Modern Techniques	POs – 1,2,3,4,5,7,9,10,11,12	PSOs – 1,2	Presentation at National Conference on Robotics, Automation, Manufacturing and Power Systems (NCRAMPS'19) Organized by the Department of Mechatronics Engineering
25.	Mechanical Operated Drip Irrigation System For Road Side Trees And Plants	POs – 1,2,3,4,5,7,9,10,11,12	PSOs – 1,2	Presentation at International Conference on recent developments in Mechanical Engineering 2019 at S.A.Engineering college, Chennai
Review/Experimental studies in emerging mechanical sciences				
26.	Effect Of Heat Treatment On Formability Of Titanium Grade 2 Alloy By Single Point Incremental Forming	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at International Conference on recent developments in Mechanical Engineering 2019 at S.A.Engineering college, Chennai
27.	Effect of Friction Stir Processing With Al ₂ O ₃ + Sic Particles on Microstructure And Hardness of AA7075 Alloy	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Experimental studies carried out on AA7075 Alloy using Friction stir process
28.	Improving Productivity Of Solar Still By Using Heat Pipes.	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2,3	Presentation at National Conference on Recent Trends in Mechanical Engineering

				(RTME 2019) Organized by IE(I) Mechanical Students Chapter and SAE India, Dr.Sivanthi Aditanar College of Engineering, Tiruchendur
29.	Pneumatic Operated Severe Plastic Deformation Using Back Pressure	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at International conference on Recent Innovation in Mechanical Engineering (ICRIME 2K19) conducted by Francis Xavier Engineering College, Tirunelveli
30.	Performance Analysis Of Solar Parabolic Trough Collector Using Phase Change Material	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2,3	Won 3 rd Prize in Science Day' 19, "Project Contest" Conducted by Ramco Institute of Technology, Rajapalayam.
31.	Numerical Simulation And Experimental Investigation Of Automobile Bumper Supporting Structure Using GFRP Composite	POs – 1,2,3,4,5,6,9,10,11,12	PSOs – 1,2,4	Presentation at National Conference on Challenges and Opportunities or Mechanical Engineering (ICOME-2019) held during 16 th March 2019 at Kongunadu College of Engineering and Technology, Thiruchirappalli
32.	Experimental Investigation Of tribological Properties On Copper Composites	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at International conference on Recent Innovation in Mechanical Engineering (ICRIME 2K19) conducted by Francis Xavier Engineering College,

				Tirunelveli
33.	Performance Analysis Of R290 As An Alternative Refrigerant To R134 In Domestic Refrigerator	POs – 1,2,3,4,5,7,9,10,11,12	PSOs – 1,3	Presentation at International Conference on Recent Innovation in Mechanical Engineering (ICRIME 2K19), Francis Xavier Engineering College, Tirunelveli
Solution to Industrial problems based Projects				
34.	Fatigue Design And Analysis Of Finished Super Heater Outlet Header Tube In Boiler Using FEA	POs -1,2,3,4,5,6,7,9,10,12	PSOs-1,2,3,4	•Bharat Heavy Electricals Limited – Trichy
35.	Reduction Of Cement Waste In Packer Section	POs -1,2,3,4,5,6,7,9,10,12	PSOs-1,2	•Ramco Cements Ltd., RR Nagar, Virudhunagar •Presentation at International conference on Recent Innovation in Mechanical Engineering (ICRIME 2K19) conducted by Francis Xavier Engineering College, Tirunelveli
36.	Conveyor For Match Inner Box	POs -1,2,3,4,5,6,7,9,10,12	PSOs-1,2	•Radha Match Company, Maipparai •Recent Trends in Engineering, Technology, Management (ICRTETM-2019), PSN College of Engineering and Technology, Tirunelveli.